

STARTUP ZONE

A 'Made In India' Solar Electric Car To Drive Sustainability

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RVCE solar car team from RV College of Engineering, Bengaluru has developed electric vehicle ARKA. The interdisciplinary team comprises more than 30 engineering students from electronics and communication, computer science, mechanical, aerospace and chemical engineering streams. It was formed in September 2013 with the motive to promote the use of solar energy in transportation sector, to drive research and innovation into solar technology and to spread awareness regarding the plausible benefits of renewable energy in our society. The team aims

to achieve high performance and build India's most efficient car. A brief description of its electronic sub-systems is given below.

Battery system (energy storage).

The first step in designing a battery pack is to map out the requirements. These are decided on the basis of optimum voltage required for operating the motor and various other loads. These also aim to maximise the number of cells used, since an increase in the number of cell units in parallel increases the current given out. In ARKA electric vehicle, the RVCE solar car team used a configuration of 29S14P, which gives an output voltage of 104V and has a rating of 47Ah.

One of the major innovations they carried out in this sub-system is the battery box. As batteries form the heart of an electric car, these need to be sturdy enough to sustain vibrations and impact. The team made the battery box out of fibre-reinforced plastic (FRP), which was CNC-machined

so that all modules could fit into it perfectly.

A modularised design allows them to replace any module, if damaged, which is not possible in conventional electric cars. Also, they use a battery management system (BMS) from Tritium, Australia. The BMS makes sure that the batteries are operating within safe limits.

The battery pack used to store energy from the solar collector must have high energy density, high number of charge-discharge cycles and must deliver high power output to drive the vehicle, and yet be lightweight. Many types of batteries can be used to store energy, but the preferred ones are lithium-ion and lithium-polymer, as these offer high energy densities for lower weight. In ARKA, Panasonic 18650B cells are used. These are also used in Tesla and other modern-day electric vehicles.

Solar array. A solar array consists of solar modules that cover the top of a solar car. It is the only source of external energy in the car that charges the batteries. Most modern-day rooftop solar panels use cells that have about 15 to 17 per cent efficiency. These use 260 SunPower Maxeon Gen 2 monocrystalline silicon cells that have an efficiency of 22.4 per cent.

Manufacturing a solar array. The manufacturing of RVCE's solar module was done at HHV Solar, Bengaluru. The encapsulation process involves sandwiching the solar cells between two layers of ethylene tetrafluoroethylene (ETFE) and joining using organic adhesive. The baseplate is made of hard glass epoxy to give mechanical strength to the array, while the top sheet is made of clear plastic to ensure that maximum light can permeate to the solar cells. Due to this innovative technique, the team manages to reduce the drop-in efficiency by only one per cent. This also makes the solar module flexible, so that it can be placed on the curved top shell of the solar car.

The area of the solar array is



Fig. 1: Solar array



Fig. 2: In-house developed battery box